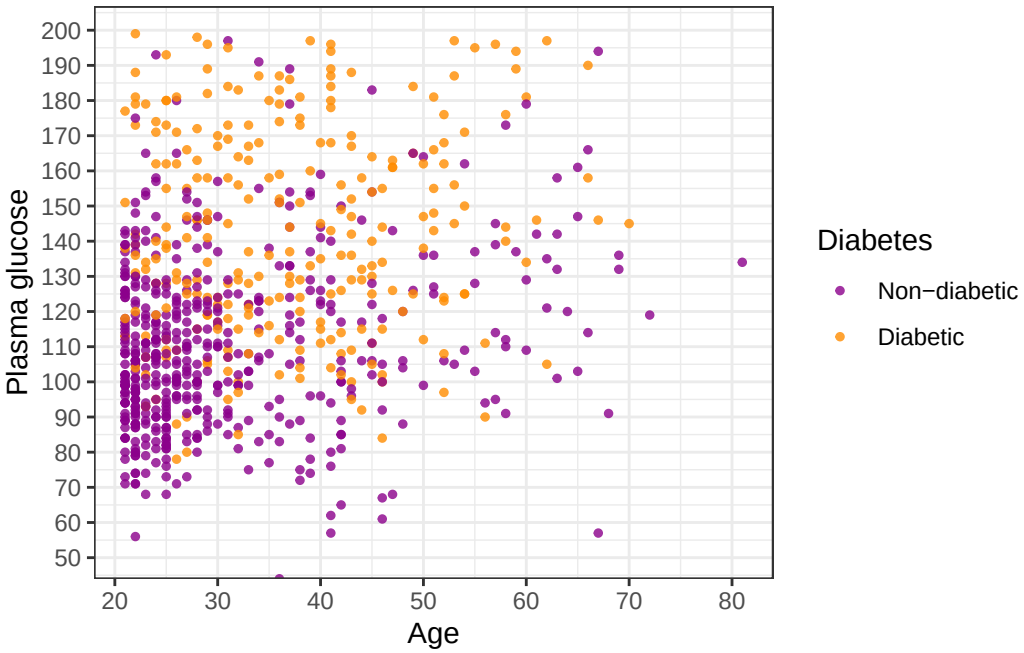
1 Assignment 3 - Logistic regression and Basis Function Expansion

Visualization of the data shows that there are values equal to zero in the measures, including plasma glucose which is one of the predictors of interest in this assignment. `Glucose == 0` is not a feasible value, but rather representing a missing value. All cases (n=5) with plasma glucose equal to zero were excluded from the data before the regression analysis.

1.1 Scatterplot diabetes vs age



Plotting plasma glucose levels vs. age shows that the two response groups, `Non-diabetic` and `Diabetic`, show a substantial overlap. `Diabetes` would therefore not be easily predicted by `Age` and `Glucose` in a standard logistic regression model which would not be able to derive a straight line clearly separating the groups.

1.2 Logistic regression model

The vector form of the probailistic equation for a logistic regression model is

$$P(Y = 1 \mid \mathbf{x}) = \text{sigmoid}(\theta^T \mathbf{x}) \tag{1}$$

where the sigmoid function is

$$sigmoid(\theta^T \mathbf{x}) = \frac{1}{1 + e^{-\theta^T \mathbf{x}}} \tag{2}$$

For `Diabetes` as the binary target/response (y), and `Glucose` and `Age` as features/predictors (x1, x2):

$$P(Y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)}} \tag{3}$$

Training a logistic regression model on the (entire) provided data set, using glm() in R, results in the following coefficients

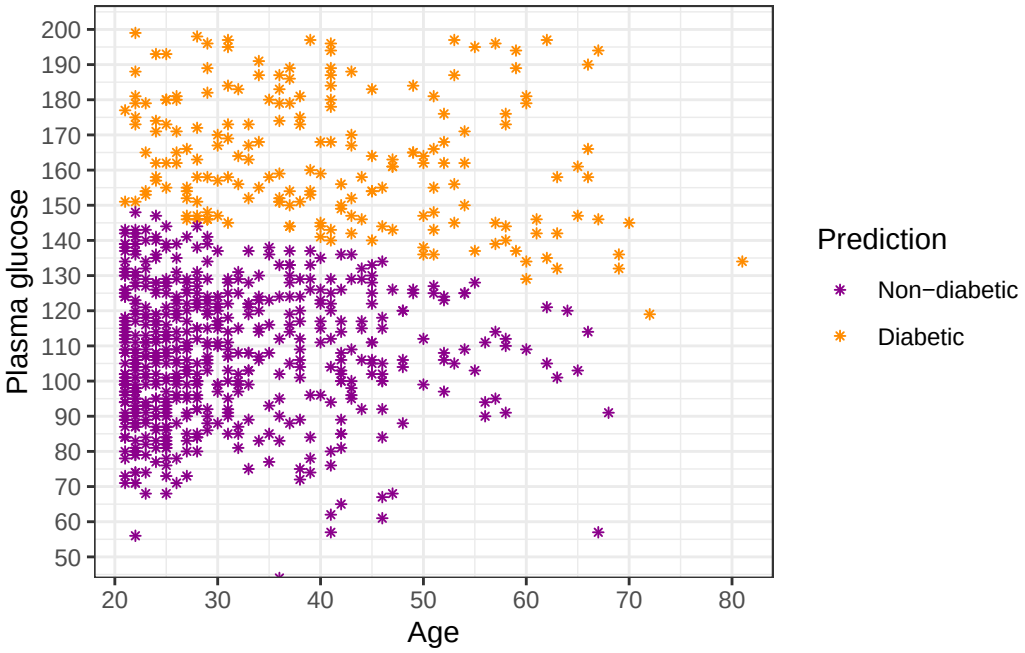
$$\theta_0 = -6.2252\theta_1 = 0.0384\theta_2 = 0.0231$$

and the corresponding probabilistic equation

$$P(Y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-6.2252+0.0384x_1+0.0231x_2)}} \tag{4}$$

The training misclassification error for this model is 0.26.

1.3 Scatterplot colored by predicions



1.4 Decision boundary equation

Setting the probabilistic equation in Equation 4 equal to `r` (here `= 0.5`) gives the approximate decision boundary equation as

$$-6.2252 + 0.0384x_1 + 0.0231x_2 = 0$$

i.e.,

$$-6.2252 + 0.0384Glucose + 0.0231Age = 0$$

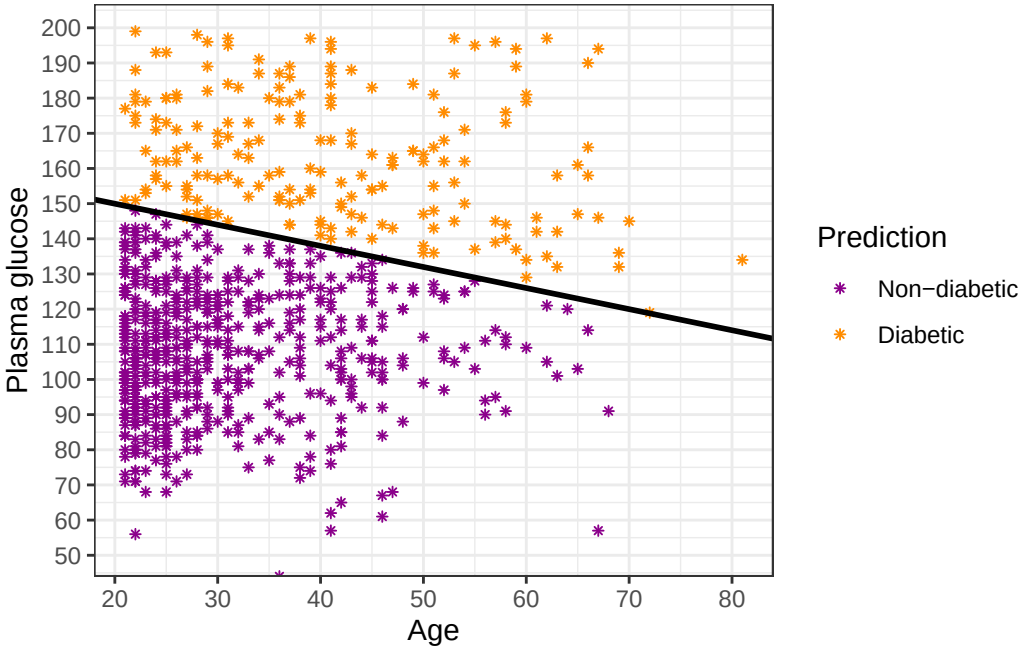
Solving this for `Glucose` gives the linear decision boundary line

$$Glucose = -\frac{\theta_0 + \theta_2 Age}{\theta_1} \tag{5}$$

approximately

$$Glucose = - - 161.97 - 0.6Age$$

Scatterplot with decision boundary line for r = 0.5



1.5 Scatterplot with decision boundary line for r = 0.2 and r = 0.8

The equation for the decision boundary line can generically be expressed as

$$x_1 = \frac{\log\left(\frac{r}{1-r}\right) - \theta_0}{\theta_1} - \frac{\theta_2}{\theta_1} x_2 \tag{6}$$

which means the intercept changes with r, the slope does not.

With r = 0.2 are more individuals predicted to be diabetic, with r increased to 0.8 instead fewer individuals. The training misclassification errors are 0.36 and 0.31 resepectively for the 0.2 and 0.8 cases, both higher than for the 0.5 cutoff case.

